Healthcare Access Under Uncertainty

Interval shortest-path bounds, executable checks and network-scale software evidence

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Engineering methods report. Real Sofala road topology; synthetic travel-time and closure inputs. No field accuracy, clinical-access validation or completed historical Idai replication.

Abstract

A travel-time map can imply more certainty than its road speeds, closures and facility records support. This project asks which threshold-based access decisions remain unchanged across an explicitly bounded family of network conditions. On a fixed directed graph with nonnegative edge-time intervals and specified facility availability, shortest paths on optimistic and pessimistic endpoint graphs provide lower and upper travel-time bounds. They yield three outcomes: definite access, definite exclusion and unresolved. The retained software evidence includes a six-node illustration, 46,656 exhaustively enumerated small graphs, 1,000,000 admissible worlds and zero recorded failures in 1,679,616 classification checks and 1,679,616 widening checks. A separate workload uses 32,577 nodes and 67,290 directed arcs from a Sofala OpenStreetMap fixture; original-plus-widened calculations took a median 139.616 ms on the recorded macOS host. Its edge costs, closures and targets are synthetic development assumptions. Thus the answer is conditional but useful: the implementation can preserve an explicit unresolved class efficiently, while the evidence does not establish real healthcare-access accuracy. The report states the mathematical guarantee, tests, computational protocol, data-qualification blockers and limits on translating software checks into geographic decisions.

Completed scope. This report closes the interval-routing method and retained software demonstration at the source revisions named below. It includes the proof, finite enumeration and the recorded Sofala-topology scaling workload. Real journey-time accuracy, service availability, clinical outcomes and historical Idai replication are outside the completed scope. No reserved independent evaluation data was opened for this closure.

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1 Central question and contribution

Question. Which access classifications remain valid when road travel times and availability vary within a declared model, and which observations would be needed before using those classifications in a real area?

Evidence-supported answer. For a fixed graph and facility set, nonnegative interval times allow exact endpoint bounds under an independent uncertainty box and conservative bounds for compatible correlated subsets. The software implements that calculation, retains ambiguity rather than forcing a binary answer, and has recorded correctness and scaling checks. The result is a statement about a modeled family of routes, not the actual service available to an individual.

This work arose in a collaborative project context with Delger. Mergen Ulziibayar's portfolio contribution is the shortest-path bounds formulation and implementation, input qualification, reproducible examples and evidence records. This does not claim sole ownership of all team work or upstream datasets. AI-assisted tools supported source/code preparation and this report; numerical claims here are traced to retained results rather than generated prose. No new reserved outcomes, field observations or simulation campaign were collected for this document.

The portfolio's frozen evidence commit is `e6a7370b4d44`; the current status review used `48bf7f5518c7`. Full identities and file hashes are retained in the accompanying source record. Numerical tables below are attached to the frozen records rather than to an unspecified moving branch [1, 2].

2 Research context and the quantity being estimated

Potential geographic access is narrower than healthcare access in practice. A network route does not establish affordability, transport availability, staffing, service capacity, whether the facility is open or whether a patient can safely make the journey. Even a correct travel-time calculation is conditional on mode, direction, time period and facility eligibility.

Petricola et al.'s Cyclone Idai study provides a relevant published development baseline for road criticality and healthcare-access loss [3]. It compares network and raster approaches, discusses incomplete OpenStreetMap mapping and assumes facilities remain operational in the absence of a reliable damage inventory. These qualifications motivate explicit uncertainty and input lineage; the present project has not reproduced that publication's national population-coverage result. A smaller, newer topology fixture cannot stand in for historical roads, facilities, flood status, population and routing configuration.

The immediate estimand is therefore deliberately simple. For origin node v, eligible facility set F, admissible world w and directed travel-time graph $G(w)$, define

$$T_w(v) = \min_{f \in F} \min_{p: v \to f} \sum_{e \in p} t_e(w). \tag{1}$$

If no eligible facility is reachable, the value is $+\infty$. A threshold τ represents the chosen driving-time criterion in minutes. It is a decision parameter, not a universal clinical standard. The toy example uses 10 min; the separate owner-selected historical reproduction target concerns a published 60 min driving quantity. Those are distinct tasks.

3 Model and assumptions

3.1 Edge uncertainty and closures

Let $G = (V, E)$ be a fixed directed graph. For a definitely available edge, travel time satisfies

$$0 \le \ell_e \le t_e(w) \le u_e \le +\infty. \tag{2}$$

An optionally closed edge is present at its lower travel time in the optimistic graph and removed in the pessimistic graph. A definitely closed edge is removed from both. This represents connectivity uncertainty explicitly rather than substituting a large arbitrary number of minutes for a closure.

For an edge of length d kilometers and positive speed bounds $v_{\min}, v_{\max}$ in km/h,

$$[\ell_e, u_e] = \left[\frac{60d}{v_{\max}}, \frac{60d}{v_{\min}}\right] \text{ minutes.} \tag{3}$$

A zero lower speed requires a closure/infinite-time interpretation. Negative values, NaNs, invalid nodes and arithmetic overflow are rejected by the implementation rather than silently accepted. None of these validation checks establishes that the chosen speed range covers real traffic conditions.

3.2 What must stay fixed

The bound proof holds for the declared topology, directionality and facility set. A missing road can make a modeled exclusion false in the real world; a missing bridge closure can make modeled access false. Adding a road or opening a facility changes the model rather than simply widening a time interval. Similarly, a flood that changes topology outside the declared optional-edge set is outside the guarantee.

The current demonstration does not implement service-capacity constraints, ambulance dispatch, transfer chains, walking-to-road access, queues or multimodal transport. A future facility-availability model could be bounded using declared optimistic/pessimistic eligible sets, but the present result fixes availability. Keeping this distinction explicit prevents the title “healthcare access” from suggesting a richer model than the implementation contains.

4 Method: lower and upper shortest-path bounds

4.1 Endpoint graphs and proof

Construct G_L with all optimistically available edges at lower times, and G_U with only definitely available edges at upper times. Let $L(v)$ and $U(v)$ be shortest travel times from v to any eligible facility on those graphs. Then

$$L(v) \le T_w(v) \le U(v) \quad \text{for every admissible } w. \tag{4}$$

For the lower inequality, every path feasible in a particular world is available in G_L with no larger edge weights; minimizing over a potentially larger path family cannot increase time. For the upper inequality, each path in G_U is available in every world with no larger actual weights. The best actual path is no worse than that pessimistic candidate. If $U = +\infty$, the upper inequality supplies no finite guarantee, as intended.

Under independent intervals and optional closures, the all-low/all-open and all-high/all-closed endpoint worlds are themselves admissible, making these extrema attainable. If physically correlated conditions restrict the family to a subset of that box, the same inequalities remain conservative; the endpoint worlds may then be unattainable. A conservative bound may leave many origins unresolved without being numerically wrong.

4.2 Classification and its boundary cases

For finite $\tau \ge 0$, the classifier is

$$C(v) = \begin{cases} \text{definite access}, & U(v) \le \tau, \\ \text{definite exclusion}, & L(v) > \tau, \\ \text{unresolved}, & \text{otherwise.} \end{cases} \tag{5}$$

Equality belongs to access. An unreachable origin in both endpoint graphs is excluded for any finite threshold. An origin connected only in some worlds is generally unresolved. The method's useful output is often this last class: it identifies a classification that the current inputs do not settle.

Widening a fixed uncertainty family can only reduce L and increase U. Consequently an already unresolved node cannot become definite merely because less is known. A definite decision may become unresolved. This monotonicity is tested explicitly; it is not an empirical geographical observation.

4.3 Efficient implementation and direction

The implementation reverses directed edges and starts a multi-source Dijkstra search from all eligible facilities. Distances in that reverse graph equal resident-to-facility times in the original graph. This matters on one-way streets: an outward facility isochrone on the original directed graph answers a different question. Nonnegative weights support Dijkstra's algorithm [4]; the production source and independent oracle are retained separately.

With a binary-heap implementation, the usual sparse-graph complexity is $O((|V| + |E|) \log |V|)$ per endpoint calculation, with storage proportional to graph and frontier size. Two endpoint searches form one bounds calculation. The recorded "paired" scaling workload performs both the original and widened calls, so it includes four endpoint searches. It should not be advertised as one single-source query, route-planning latency or end-to-end remote-data ingestion time.

5 Retained computational experiments

5.1 Protocol and comparison logic

The computational evidence has three complementary layers. The worked example makes the classification semantics readable. Exhaustive finite enumeration checks implementation behavior against admissible worlds on small graphs. A real-topology fixture exercises larger sparse networks and compares endpoint distances to an independent routing calculation. These layers test different failure modes; none supplies field validation.

The primary hypotheses are software properties: no declared definite classification contradicts any enumerated admissible world; widening preserves the monotonicity rule; production endpoint distances agree with an independent oracle on the retained workloads. Zero failures means the recorded checks passed. It does not assign a probability that all possible programs, floating-point cases or real-world inputs are correct.

5.2 Six-node demonstration

Table 1 reproduces the exact saved example. It includes a permanently disconnected origin, an optional connection and a facility node. Widening retains both certain and uncertain outcomes in the expected direction.

5.3 Exhaustive software checks

The committed result records 46,656 graphs and 1,000,000 worlds, with 1,679,616 classification checks and the same number of widening checks, and zero failures. The original execution records Python 3.11.5 and hashes of the demo, runner, implementation and tests. Finite enumeration corroborates the implementation for that finite family. The guarantee over continuous intervals comes from the monotonic proof in Section 4, not from pretending that a million discrete worlds cover the continuum.

Table 1: Saved example at $\tau = 10$ min. Bounds in minutes [1].

	Original family			Widened family	
Node	L	U	Class	$[L, U]$	Class
A	5	8	Access	$[3, 12]$	Unresolved
B	18	25	Exclusion	$[8, 30]$	Unresolved
C	7	11	Unresolved	$[4, 17]$	Unresolved
D	4	∞	Unresolved	$[2, \infty]$	Unresolved
E	∞	∞	Exclusion	$[\infty, \infty]$	Exclusion
Clinic	0	0	Access	$[0, 0]$	Access

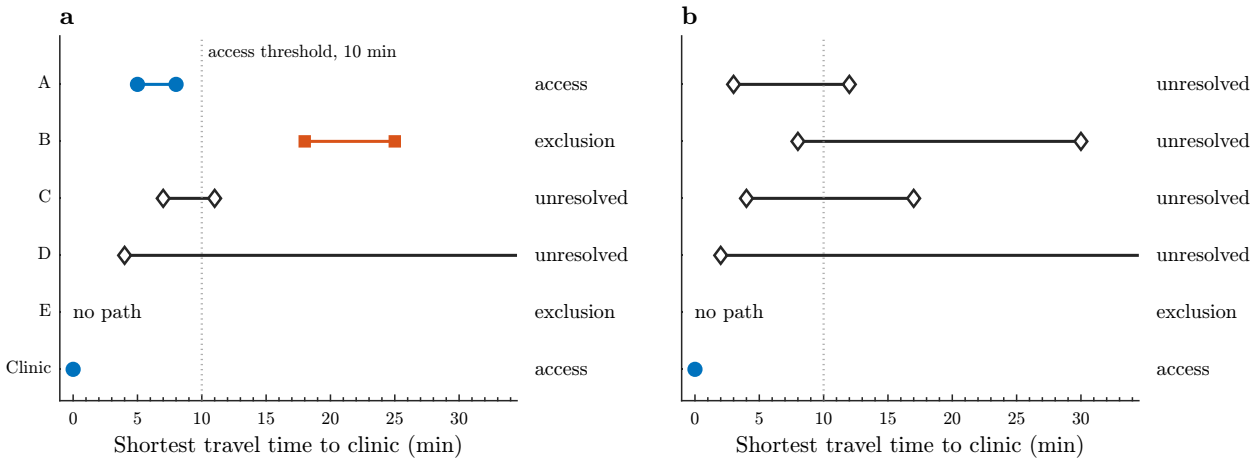


Figure 1: Retained toy-network figure, reproduced from the project's vector output. These synthetic origin labels and bounds illustrate the three-way classifier; they are not settlements or observed patient travel times. Credit: Mergen Ulziibayar, healthcare-access-uncertainty [1].

The saved result also records a speed-conversion example with bounds $[10, 30]$ minutes. This is a unit and endpoint check of Equation 3. It does not estimate a travel-speed distribution or give a confidence interval.

5.4 Sofala-topology scaling workload

The larger workload uses a retained Beira-area/Sofala OpenStreetMap graph, with synthetic time and closure assumptions, fixed synthetic targets, and seed 61011. Table 2 shows the retained five-repeat timings. These are not newly benchmarked values on this Windows machine.

Table 2: Recorded original-plus-widened bounds workload. Ranges are observed minima and maxima over five repetitions, not confidence intervals [1].

Nodes	Directed arcs	Median (ms)	Range (ms)	Peak RSS (MiB)	Error (min)
128	243	0.280	0.277–0.344	131.8	0
512	987	1.248	1.221–1.334	132.0	0
2,048	4,193	5.446	5.397–5.533	133.0	0
8,192	16,848	21.718	21.601–55.892	142.0	0
32,577	67,290	139.616	97.509–168.787	170.7	0

The host record is macOS 14.7.3 on arm64, Python 3.11.5, SciPy 1.17.1 and NumPy 2.4.6. Peak RSS is process high-water memory including imports, the full fixture, oracle and preceding graph sizes. Thus

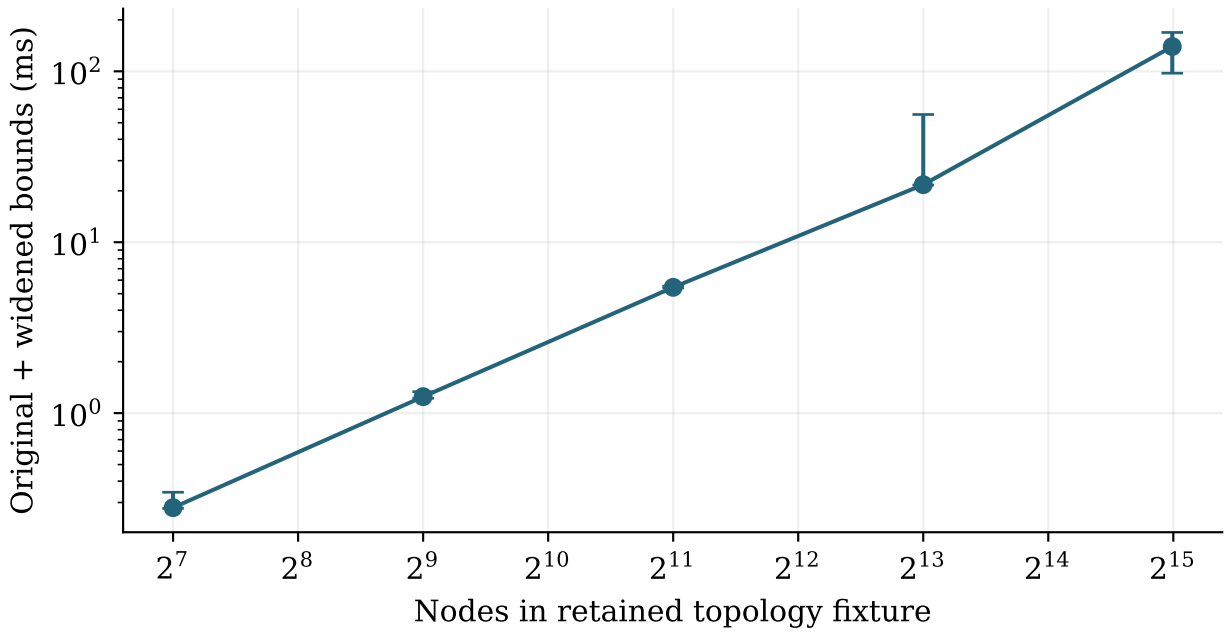


Figure 2: Replot of the committed scaling table. Markers show five-run medians; whiskers show the observed min–max range. Both axes are logarithmic. Network topology contains © OpenStreetMap contributors data under ODbL 1.0 [6]; costs and closures are synthetic. The figure describes a computational workload, not real access accuracy.

it cannot be interpreted as the incremental storage needed for each row. The software record reports zero endpoint error against its oracle and zero widening failures at each size; ordinary floating-point agreement does not establish a rigorous error bound near every possible decision threshold.

The full fixture includes 17 weak and 25 strong components. The retained lower graph has 3,113 unreachable nodes and the upper graph 29,782. This large pessimistic unreachable set is a feature of the synthetic closure setup, not evidence that almost all real Sofala residents lose access. Sampling and clipping a graph can also disconnect otherwise connected roads. The source record keeps topology identity, skipped features and fixture bounds alongside timings for that reason.

5.5 Transfer to fixed-graph path planning

The endpoint construction is not specific to healthcare semantics. Replacing facilities by a fixed robot goal set yields the same shortest-path formulation when edge costs are additive and nonnegative, topology/availability rules are declared, and uncertainty is represented by the same compatible interval family. This is a mathematical transfer of Eqs. 4–5, not a new robot experiment. It does not supply collision checking, dynamically changing obstacles, kinodynamic feasibility or closed-loop control.

6 Uncertainty and what the experiments do not identify

6.1 Input uncertainty versus numerical error

The interval box describes assumed epistemic uncertainty. It does not assign a probability to each admissible world. An unresolved result therefore is not “50% likely to have access”, and a definite result is not an empirical 100% reliability estimate. Its guarantee holds only when the true state lies inside the declared family.

The interval width $U - L$ can identify where the model remains ambiguous, but it is not by itself the value of a new road observation. A measurement must change a particular edge/facility constraint and then alter a relevant decision. Correlated flooding, uncertain transport mode and missing roads can dominate the uncertainty even when individual speed intervals appear narrow. A future observation-selection policy should compare decision improvement against acquisition cost rather than minimize interval width without a decision objective.

Floating-point calculations introduce a separate issue. Near τ, accumulated rounding or data-conversion error can affect comparison. The proof is exact over real numbers; the implementation uses normal finite arithmetic. A certified near-threshold decision would need directed rounding or a separately justified numerical tolerance. The reported oracle agreement is a useful implementation check, not that certification.

6.2 No country-level or clinical accuracy claim

Real topology does not turn synthetic costs into real travel times. The project has no independently validated distribution of road speed, flood passability or facility service status for this fixture. It has not shown classification accuracy against observed journeys, clinical outcomes, facility attendance or population-level access. Timings cannot supply any of these missing comparisons.

The retained UCI trip parser demonstration found 19 usable trips in a non-Mongolian dataset. That establishes a limited parsing/route-feasibility example, not validation of Mongolian roads or off-road access. The earlier detector track remains preserved but inactive; it was superseded as a project direction, rather than scientifically refuted by the routing result.

7 Historical replication and independent evaluation plan

The adopted historical target is a published potential-access quantity, not all conclusions of the Idai paper. Reproduction requires the correct denominator and partition, national population cells, facility set, road vintage, network profile, flood membership, and route direction. Published rounded time bands alone do not define a numerical tolerance for population-coverage error. The development record identifies discrepancies and missing identities rather than declaring replication complete.

Table 3: Inputs that must be qualified before a real-area comparison.

Input	Current evidence issue	Required resolution
Road network	Later OSM exports and ambiguous historical source dates	Pin exact extraction, geometry, tags and permitted reuse; document retrospective assumptions.
Facilities	Export identity, availability and service eligibility unresolved	Fixed, dated eligible facility set; operational-status basis.
Population	Coverage depends on denominator and cell allocation	Pin raster/version, nodata rule and fractional overlap treatment.
Routing	Direction and routing profile/build can change reachability	Explicit resident-to-facility rule, mode and exact profile/software.
Flood/closures	Membership and operational meaning uncertain	Independently sourced, dated passability constraints rather than visual flood presence alone.

Input	Current evidence issue	Required resolution
Evaluation observations	WFP candidate remains unopened and ineligible	Dates, vehicle scope, independent lineage and a preregistered row-level use rule before access.

An honest future evaluation would freeze development assumptions, identify eligible independent observations, register metrics and exclusions, and only then open reserved outcomes. It should report both the fraction of definite decisions and their observed contradiction rate within the evaluation scope. A system that declares nearly every origin unresolved may be conservative but of little operational value. A system with high definite coverage can be useful only if the family was justified independently rather than tightened until it agrees with outcomes.

No such independent evaluation was executed for this report. The proposed procedure is a way to make future evidence interpretable, not a backfilled protocol for results already observed.

8 Conclusions

1. Optimistic and pessimistic endpoint searches bound every admissible shortest travel time on the declared fixed graph and facility set (Eq. 4). Threshold classification retains an explicit unresolved outcome rather than assigning an unsupported binary label (Eq. 5, Table 1).

2. The retained finite software experiment reports 46,656 graphs, 1,000,000 worlds and zero failures across 1,679,616 classification checks and the same number of widening checks (Section 5.3). The continuous-interval statement follows from the proof, not from pretending enumeration covers every real input.

3. The 32,577-node, 67,290-arc Sofala-topology fixture ran the original-plus-widened workload in a recorded five-repeat median of 139.616 ms (Table 2). Those costs, closures and targets are synthetic, and the timing is tied to its recorded host.

4. The same mathematical construction transfers to fixed-graph path planning with additive nonnegative uncertain edge costs and a fixed goal set (Section 5.5). No robot execution, dynamic obstacle handling or control-loop performance was evaluated.

5. The completed result is an uncertainty-handling software method. Real-area healthcare accuracy and historical replication remain unevaluated because input identity and independent observations are unresolved. The engineering recommendation is to qualify the graph, costs, closures and facility eligibility before interpreting modeled classes operationally.

9 Future work

1. First perform the cheapest empirical falsification: collect a small, independently timed route sample with dated passability observations, then check whether declared intervals and route directions cover it. This is a proposed field protocol, not an executed or authorized campaign.

2. Qualify one dated road/facility/population/closure dataset and freeze evaluation rules before opening reserved outcomes; assess definite-decision coverage and contradiction rate together.

3. Evaluate a fixed-graph robotics-planning transfer separately, including near-threshold numerical treatment and the cost of additional edge observations. Keep dynamic obstacles and closed-loop execution outside its claims until tested.

A Reproducibility record and reuse terms

The complete provenance file accompanying the report contains full commits, hashes, source paths and numerical claim mappings. The compact map below identifies the authoritative numerical homes.

Record	Purpose
`analysis/catanroads/access_bounds.py`	Endpoint bounds and input validation.
`analysis/run_access_demo.py`	Finite demonstration and exhaustive checks.
`evidence/access-stability-20261006/result.json`	Exact toy results and enumeration counts.
`evidence/sofala-scaling/scaling.json`	Five-repeat timings, environment, graph hash and oracle checks.
`docs/ACCESS_STABILITY.md`	Mathematical assumptions and reproduction target.
`results/access-stability/bounds.pdf`	Retained vector figure reproduced here.

The repository documents an offline figure-reproduction path using committed outputs. This report copied the saved bounds figure and replotted the saved timing table; it did not rerun a study, acquire remote data or open reserved evaluation outcomes. Production code is MIT licensed. The isolated upstream compatibility harness has GPL-3.0-only terms and is not incorporated into production. Third-party inputs retain their own conditions. The topology is credited to © OpenStreetMap contributors and available under ODbL 1.0 [6]. No Idai map image or third-party photograph is reproduced.

B Worked boundary checks for interpretation

At $\tau = 10$, an interval $[5, 10]$ means definite access; $[10, 11]$ is unresolved because some admissible times exceed the threshold; $[10.1, 12]$ is excluded. These are algebraic illustrations of Equation 5, not additional executed cases. An edge closure can increase the upper bound to infinity without changing the lower bound. If an entire origin has $[\infty, \infty]$, exclusion follows inside the graph model, but an unmapped real road would invalidate that model premise. These cases explain why a precise class label still needs a precise domain of validity.

References

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[2] M. Ulziibayar and project contributors. *Healthcare access under uncertainty: current status reviewed for this report*. Commit `48bf7f5518c71a8afb361a6ee46e7577fd06a4aa`. `https://github.com/500ft/healthcare-access-uncertainty/tree/48bf7f5518c71a8afb361a6ee46e7577fd06a4aa`.

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[4] E. W. Dijkstra. A note on two problems in connexion with graphs. *Numerische Mathematik*, 1:269–271, 1959. `https://doi.org/10.1007/BF01386390`.

[5] SciPy developers. *scipy.sparse.csgraph.dijkstra*: nonnegative directed shortest paths and multi-source minimum-distance interface. Official documentation consulted 11 November 2026; the retained benchmark used SciPy 1.17.1. `https://docs.scipy.org/doc/scipy/reference/generated/scipy.sparse.csgraph.dijkstra.html`.

[6] OpenStreetMap Foundation. *Copyright and License*. OpenStreetMap contributors; ODbL 1.0. `https://www.openstreetmap.org/copyright`.